

Chapter 4: Method of Frobenius

Special Functions of Mathematical Physics: A Tourist's Guidebook

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July 5, 2026

- 1 The guitar string and the drum head
- 2 When will the method of Frobenius work?
- 3 Second solutions

Where we are going

Separation of variables turned our PDEs into linear, second-order, homogeneous ordinary differential equations. Two of them recur:

- the **guitar string** gave the harmonic oscillator equation

$$\frac{d^2 y}{dx^2} + k^2 y = 0,$$

- the **drum** gave **Bessel's equation**

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (k^2 x^2 - m^2) y = 0.$$

This chapter develops one general method for the whole class

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0.$$

Second-order means we have a second derivative of y . *Linear* means P and Q depend only on x , not on y . *Homogeneous* means the right-hand side is zero. Just as analytic and meromorphic functions had Taylor and Laurent representations, we will solve these ODEs with **series**. That technique is the *method of Frobenius*.

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Guitar string: the harmonic oscillator by Taylor series

Solve the harmonic oscillator (here k has been rescaled into x , i.e. $kx \mapsto x$):

$$\frac{d^2 y}{dx^2} + y = 0, \quad y(0) = 0, \quad y(\pi) = 0. \quad (4.1.1)$$

If the solution is analytic it has a Taylor series

$$y(x) = \sum_{n=0}^{\infty} a_n x^n,$$

and the whole task is to find the coefficients a_n . Substitute into (4.1.1):

$$\frac{d^2}{dx^2} \left(\sum_{n=0}^{\infty} a_n x^n \right) + \left(\sum_{n=0}^{\infty} a_n x^n \right) = 0.$$

Differentiating term by term, the first sum loses its $n = 0, 1$ terms (they are annihilated by the second derivative), so it starts at $n = 2$:

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0.$$

Guitar string: the harmonic oscillator by Taylor series (cont.)

To combine the sums we line up the powers of x . Replace $n \mapsto n + 2$ in the first sum; its lower limit moves from $n = 2$ to $n = 0$ and the upper limit stays ∞ :

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=0}^{\infty} a_n x^n = 0.$$

Now both sums carry x^n , so match coefficients:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n] x^n = 0.$$

This holds for all x only if every bracket vanishes:

$$(n+2)(n+1)a_{n+2} + a_n = 0 \quad \implies \quad a_{n+2} = -\frac{1}{(n+1)(n+2)} a_n.$$

The recurrence skips every other term: a_0 feeds $a_2, a_4, \dots$ and a_1 feeds $a_3, a_5, \dots$

Resolving the recurrence into $\cos x$ and $\sin x$

Starting from a_0 (even index), each step multiplies by $-1/[(2j-1)(2j)]$:

$$a_2 = -\frac{1}{1 \cdot 2} a_0 = -\frac{1}{2!} a_0, \quad a_4 = -\frac{1}{3 \cdot 4} a_2 = \frac{1}{4!} a_0, \quad \dots$$

Each new step inserts the next two integers into the denominator, so out to a_{2n} the denominator builds up the consecutive product $1 \cdot 2 \cdot 3 \cdots (2n) = (2n)!$ and the sign accumulates n factors of -1 :

$$a_{2n} = (-1)^n \frac{1}{(2n)!} a_0.$$

Starting from a_1 (odd index), each step multiplies by $-1/[(2j)(2j+1)]$:

$$a_3 = -\frac{1}{2 \cdot 3} a_1, \quad a_5 = -\frac{1}{4 \cdot 5} a_3 = \frac{1}{2 \cdot 3 \cdot 4 \cdot 5} a_1 = \frac{1}{5!} a_1, \quad \dots$$

Here the denominator runs $2 \cdot 3 \cdots (2n+1) = (2n+1)!$, so

$$a_{2n+1} = (-1)^n \frac{1}{(2n+1)!} a_1.$$

Hence the general solution of (4.1.1) is

$$y(x) = a_0 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} + a_1 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

Resolving the recurrence into $\cos x$ and $\sin x$ (cont.)

These two Taylor series are exactly $\cos x$ and $\sin x$:

$$y(x) = a_0 \cos x + a_1 \sin x.$$

The boundary conditions $y(0) = 0$ and $y(\pi) = 0$ force $a_0 = 0$, leaving the string's mode shape $y(x) = a_1 \sin x$.

Drum: Bessel's equation needs a generalized series

Now solve Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - m^2) y = 0. \quad (4.1.2)$$

For the drum m is a nonnegative integer. The equation makes sense for any m , but in this Frobenius derivation we keep the drum case $m = 0, 1, 2, \dots$. We cannot assume a plain Taylor series: the solution may have a singularity at $x = 0$. So we allow a fractional/negative leading power r (with $a_0 \neq 0$):

$$y(x) = x^r \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+r}. \quad (4.1.3)$$

Substituting (4.1.3) into (4.1.2) and differentiating each series,

$$x^2 \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2} + x \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1} + (x^2 - m^2) \sum_{k=0}^{\infty} a_k x^{k+r} = 0.$$

Drum: Bessel's equation needs a generalized series (cont.)

Because r is still unknown, no term is guaranteed to drop, so we keep every sum starting at $k = 0$. Distributing x^2 , x , and $(x^2 - m^2)$:

$$\sum a_k(k+r)(k+r-1)x^{k+r} + \sum a_k(k+r)x^{k+r} + \sum a_k x^{k+r+2} - \sum a_k m^2 x^{k+r} = 0.$$

Drum: Bessel's equation needs a generalized series (cont.)

In the third sum (the only one with x^{k+r+2}) set the new index $j = k + 2$, so $k = j - 2$ and the term $a_k x^{k+r+2}$ becomes $a_{j-2} x^{j+r}$; renaming j back to k gives $\sum_{k=2}^{\infty} a_{k-2} x^{k+r}$. Extending it down to $k = 0$ with the bookkeeping $a_{-2} = a_{-1} = 0$ (those coefficients do not exist), all four sums now share the power x^{k+r} and the same lower limit:

$$\sum_{k=0}^{\infty} \left(a_k(k+r)(k+r-1) + a_k(k+r) + a_{k-2} - a_k m^2 \right) x^{k+r} = 0.$$

Collect the a_k terms. Since $(k+r)(k+r-1) + (k+r) = (k+r)^2$,

$$\left[(k+r)^2 - m^2 \right] a_k + a_{k-2} = 0. \quad (4.1.4)$$

At $k = 0$, with $a_{-2} = 0$ and $a_0 \neq 0$, the bracket itself must vanish:

$$r^2 - m^2 = 0. \quad (4.1.5)$$

This is the *indicial equation*; it fixes the leading exponent, $r = \pm m$. Take $r = +m$ with $m \geq 0$ so the solution stays finite at $x = 0$.

The Bessel recurrence and $J_m(x)$

With $r = m$, the bracket in (4.1.4) becomes $(k + m)^2 - m^2 = k(k + 2m)$, so

$$k(k + 2m) a_k = -a_{k-2}. \quad (4.1.6)$$

At $k = 1$ this reads $(1 + 2m)a_1 = 0$. Since $m \geq 0$ (so $1 + 2m \neq 0$), we get $a_1 = 0$, and therefore every odd coefficient vanishes. For the even ones,

$$a_k = \frac{-1}{k(k + 2m)} a_{k-2}.$$

Iterating from a_0 , and writing each new denominator $k(k + 2m)$ with $k = 2, 4, \dots$:

$$\begin{aligned} a_2 &= \frac{-1}{2(2 + 2m)} a_0 = \frac{-1}{4(1 + m)} a_0, \\ a_4 &= \frac{-1}{4(4 + 2m)} a_2 = \frac{1}{8 \cdot 4(2 + m)(1 + m)} a_0, \\ a_6 &= \frac{-1}{6(6 + 2m)} a_4 = -\frac{1}{4^3 3! (m + 3)(m + 2)(m + 1)} a_0. \end{aligned}$$

and the odd coefficients $a_3 = a_5 = \dots = 0$ since $a_1 = 0$. At step j the recurrence contributes a factor $-1/[(2j)(2j + 2m)] = -1/[4j(j + m)]$. Multiplying these j factors

The Bessel recurrence and $J_m(x)$ (cont.)

collects a $(-1)^j$, a 4^{-j} , a $j! = 1 \cdot 2 \cdots j$ from the first factors, and $(m+1)(m+2) \cdots (m+j) = (m+j)!/m!$ from the second:

$$a_{2j} = (-1)^j \frac{m!}{2^{2j} j! (m+j)!} a_0, \quad a_{2j+1} = 0. \quad (4.1.7)$$

The Bessel recurrence and $J_m(x)$ (cont.)

So the series solution (4.1.3), summing $a_{2j}x^{2j+m}$, is

$$y(x) = a_0 m! \sum_{j=0}^{\infty} \frac{(-1)^j}{2^{2j} j! (m+j)!} x^{2j+m}.$$

Pull the powers of two into a half-argument: $x^{2j+m}/2^{2j} = 2^m (x/2)^{2j+m}$, so

$$y(x) = a_0 2^m m! \sum_{j=0}^{\infty} \frac{(-1)^j}{j! (m+j)!} \left(\frac{x}{2}\right)^{2j+m}.$$

Defining the **Bessel function of the first kind** of order m ,

$$J_m(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (m+k)!} \left(\frac{x}{2}\right)^{2k+m}, \quad (4.1.8)$$

the solution reads $y(x) = a_0 2^m m! J_m(x)$.

Fixing the constant. For $m = 0, 1, 2, \dots$, the leading coefficient a_0 was free; the standard choice $a_0 = 1/(2^m m!)$ cancels the prefactor $2^m m!$ and leaves exactly

The Bessel recurrence and $J_m(x)$ (cont.)

$y(x) = J_m(x)$. (This is the conventional normalization, fixing J_m so that its leading term is $(x/2)^m/m!$.)

For an integer order $-m$ ($m > 0$) the method fails: the recurrence

$a_k = -a_{k-2}/[k(k-2m)]$ blows up at $k = 2m$, where the denominator hits zero.

Frobenius does not always deliver both solutions; Bessel's second solution waits for §4.3.

What the other Bessel root is trying to say

Bessel's equation contains m^2 , so it is often cleaner to write the order as ν and remember that the indicial roots are

$$r = \nu \quad \text{and} \quad r = -\nu.$$

The root $r = \nu$ gave the regular solution. With the conventional normalization, and with factorials replaced by gamma functions when ν is not an integer, it is

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}.$$

If $\nu \notin \mathbb{Z}$, the other root gives the companion Frobenius solution

$$J_{-\nu}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k - \nu + 1)} \left(\frac{x}{2}\right)^{2k-\nu}.$$

For noninteger ν , these two leading powers x^ν and $x^{-\nu}$ are different enough to make the solutions independent.

What the other Bessel root is trying to say (cont.)

For order $\nu = \frac{1}{2}$, the root $r = -\frac{1}{2}$ gives the companion half-order solution in its simplest form. Choose that root. Then

$$\left[\left(k - \frac{1}{2}\right)^2 - \frac{1}{4}\right] a_k + a_{k-2} = 0 \quad \Longleftrightarrow \quad k(k-1)a_k = -a_{k-2}.$$

The $k = 1$ equation is $0 \cdot a_1 = 0$, so a_1 is free instead of being forced to vanish. The two chains are

$$a_{2j} = (-1)^j \frac{a_0}{(2j)!}, \quad a_{2j+1} = (-1)^j \frac{a_1}{(2j+1)!}.$$

Thus

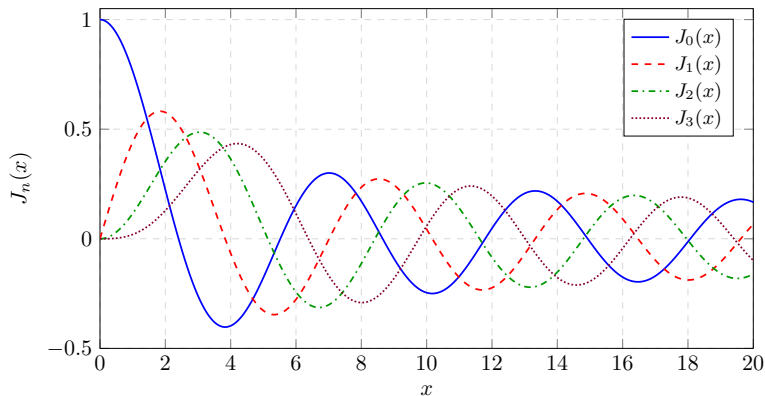
$$y(x) = x^{-1/2} (a_0 \cos x + a_1 \sin x).$$

After standard normalization,

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x, \quad J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x.$$

For positive integer order, the negative-root branch is the one that breaks; the missing independent solution is the singular Y_m constructed later by reduction of order.

The Bessel functions of the first kind $J_0, \dots, J_3$



Near the origin, $J_0(0) = 1$ while $J_m(x)$ for $m \geq 1$ first behaves like a constant multiple of x^m . Farther out the Bessel functions oscillate with a slowly decaying amplitude. The zeros of J_m are the allowed radial constants k that clamp the drum's rim.

The method of Frobenius: the recipe

To solve $p_0(x)y'' + p_1(x)y' + p_2(x)y = 0$:

- 1 Substitute the generalized series $y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$.
- 2 Shift the indices so all powers of x are the same.
- 3 Combine the matching coefficients together.
- 4 Solve the *indicial equation* (the $k = 0$ coefficient) to determine r .
- 5 Develop a recursion formula starting with a_0 .

The leading exponent r comes from the lowest power of x ; once r is fixed, the recurrence generates all higher coefficients from a_0 (and, when the recurrence allows, a_1).

Example: Airy's equation by series

Airy's equation

$$y'' - xy = 0 \quad (4.1.9)$$

arises for a quantum particle in a constant force (linear potential) and in the diffraction of light. Its behavior is already readable off the equation: for $x < 0$ it looks like $y'' + \lambda^2 y = 0$ (oscillatory, $\sin \lambda x, \cos \lambda x$), and for $x > 0$ like $y'' - \lambda^2 y = 0$ (exponential, $e^{\pm \lambda x}$).

Take $y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}$, so $y'' = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2}$, and (4.1.9) becomes

$$\sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2} - \sum_{k=0}^{\infty} a_k x^{k+r+1} = 0.$$

Align powers: in the second sum the power is x^{k+r+1} , and setting the new index $j = k + 3$ (so $k = j - 3$) turns $a_k x^{k+r+1}$ into $a_{j-3} x^{j+r-2}$; renaming $j \rightarrow k$ and using $a_{-3} = a_{-2} = a_{-1} = 0$ to start at $k = 0$,

$$\sum_{k=0}^{\infty} (a_k (k+r)(k+r-1) - a_{k-3}) x^{k+r-2} = 0.$$

Example: Airy's equation by series (cont.)

Hence the coefficient of each power vanishes:

$$a_k(k+r)(k+r-1) - a_{k-3} = 0.$$

The $k=0$ term is the indicial equation $r(r-1)=0$, so $r=0$ or $r=1$. Here it does not matter which we take; both give a Taylor series ($x=0$ is an ordinary point). Take $r=0$. Then $a_k(k)(k-1) - a_{k-3} = 0$; replacing k by $k+3$ (so that $(k+3)(k+3-1) = (k+3)(k+2)$) puts the recurrence in forward form:

$$k(k-1)a_k - a_{k-3} = 0 \quad \implies \quad a_{k+3} = \frac{1}{(k+3)(k+2)} a_k.$$

Example: Airy's equation by series (cont.)

The index jumps by 3, so a_0, a_1, a_2 seed three residue-class chains. At $k = 2$ the relation reads $2(1)a_2 - a_{-1} = 0$, and with $a_{-1} = 0$ this forces $a_2 = 0$, killing the a_{3k+2} chain. From a_0 (each step multiplies by $1/[(3j)(3j-1)]$):

$$\begin{aligned}a_3 &= \frac{1}{3 \cdot 2} a_0, \\a_6 &= \frac{1}{6 \cdot 5} a_3 = \frac{1}{6 \cdot 5 \cdot 3 \cdot 2} a_0, \\a_9 &= \frac{1}{(9 \cdot 8)(6 \cdot 5)(3 \cdot 2)} a_0, \\a_{3k} &= \frac{(3k-2) \cdot (3k-5) \cdots 1}{(3k)!} a_0.\end{aligned}$$

The numerator is a product stepping down by 3. Writing the *multifactorial* $n!^{(j)} = n \cdot (n-j)!^{(j)}$ with $n!^{(j)} = 1$ for $-j < n \leq 0$, this is $(3k-2)!!!$ (triple factorial), so

$$a_{3k} = \frac{(3k-2)!!!}{(3k)!} a_0.$$

Example: Airy's equation by series (cont.)

The a_1 chain works the same way, shifted by one power:

$$a_4 = \frac{1}{4 \cdot 3} a_1,$$

$$a_7 = \frac{1}{7 \cdot 6} a_4 = \frac{1}{7 \cdot 6 \cdot 4 \cdot 3} a_1,$$

$$a_{10} = \frac{8!!!}{10!} a_1,$$

$$a_{3k+1} = \frac{(3k-1)!!!}{(3k+1)!} a_1.$$

$$\begin{aligned} y(x) = & a_0 \sum_{k=0}^{\infty} \frac{(3k-2)!!!}{(3k)!} x^{3k} \\ & + a_1 \sum_{k=0}^{\infty} \frac{(3k-1)!!!}{(3k+1)!} x^{3k+1}. \end{aligned} \tag{4.1.10}$$

Mathematical detour: Airy via Fourier transform

Airy's equation can also be solved by transforming it. The calculation below is a formal Fourier-transform derivation: one can justify the integrations by parts and inversion by temporarily damping the oscillatory integrals and then taking the damping to 0^+ , or by working in the distribution sense. With

$$\hat{y}(t) = F\{y(x)\}(t) = \int_{-\infty}^{\infty} y(x) e^{-itx} dx, \quad y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{y}(t) e^{itx} dt.$$

Integration by parts gives the derivative rule

$$F\{y'\} = it F\{y\}, \quad \text{repeating once:} \quad F\{y''\} = -t^2 F\{y\}.$$

For multiplication by x , differentiate $\hat{y}(t) = \int y e^{-itx} dx$ under the integral:

$$\hat{y}'(t) = \int y (-ix) e^{-itx} dx = -i F\{xy\} \implies F\{xy\} = +i (F\{y\})'.$$

Mathematical detour: Airy via Fourier transform (cont.)

Transforming $y'' - xy = 0$,

$$F\{y'' - xy\} = F\{y''\} - F\{xy\} = -t^2 \hat{y} - i \hat{y}' = 0.$$

This is separable: $-i \hat{y}' = t^2 \hat{y}$, i.e.

$$\frac{\hat{y}'}{\hat{y}} = it^2 \implies \hat{y} = c e^{it^3/3}.$$

(Using $1/(-i) = i$.) Inverting,

$$y(x) = \frac{c}{2\pi} \int_{-\infty}^{\infty} e^{it^3/3} e^{itx} dt = \frac{c}{2\pi} \int_{-\infty}^{\infty} e^{i(t^3/3 + tx)} dt = \frac{c}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + tx\right) dt,$$

the last step keeping the even (cosine) part since the odd part integrates to zero. We name

$$Ai(x) = \frac{1}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + xt\right) dt, \quad (4.1.11)$$

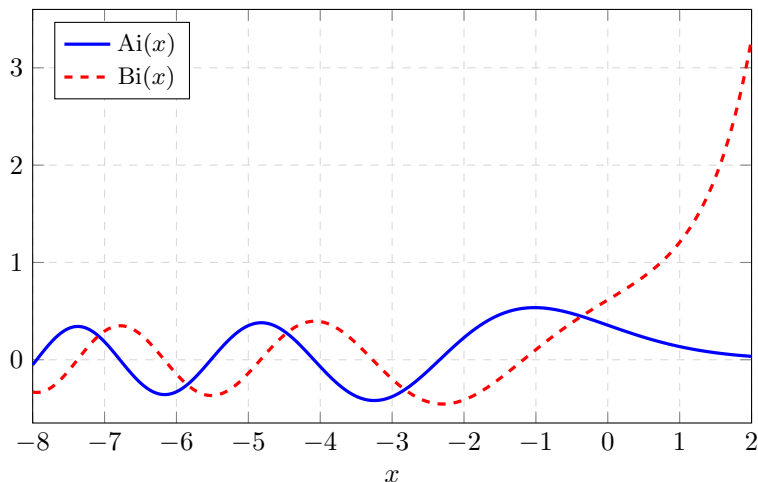
Mathematical detour: Airy via Fourier transform (cont.)

the **Airy function of the first kind**, and the second kind

$$Bi(x) = \frac{1}{\pi} \int_0^\infty \left[\exp\left(-\frac{t^3}{3} + xt\right) + \sin\left(\frac{t^3}{3} + xt\right) \right] dt. \quad (4.1.12)$$

The real-line Fourier transform selected one particular solution, Ai ; Bi is the other standard independent Airy solution and is introduced here for comparison. A known value (from the gamma function) is $Ai(0) = (3^{2/3} \Gamma(2/3))^{-1}$.

The Airy functions Ai and Bi



For $x < 0$ both standard Airy solutions oscillate. For $x > 0$, $Ai(x)$ decays while $Bi(x)$ grows, so boundary conditions usually decide which combination is physically allowed.

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Classifying the point x_0

For $y'' + P(x)y' + Q(x)y = 0$, classify the point x_0 by the coefficients P and Q :

- x_0 is an **ordinary point** if P and Q are analytic at x_0 .
- x_0 is a **regular singular point** if x_0 is singular and $(x - x_0)P(x)$ and $(x - x_0)^2 Q(x)$ are analytic at x_0 .
- Otherwise x_0 is an **irregular singular point**.

Classifying the point x_0 (cont.)

The behavior of P and Q predicts the outcome.

- 1 If both $P(x)$ and $Q(x)$ are analytic near x_0 (in particular, no pole there), then x_0 is an *ordinary point*. The equation has *two* distinct series solutions; the radius of convergence is at least the smaller of the radii of P and Q .
- 2 **Fuchs' theorem.** If x_0 is singular, P has at most a simple pole, and Q has at most a double pole at x_0 —i.e. $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0 —then x_0 is a *regular singular point*, and there is at least one series solution

$$y = \sum_{k=0}^{\infty} a_k (x - x_0)^{k+r}.$$

If the two roots of the indicial equation differ by a noninteger, there are two independent Frobenius solutions. If the roots coincide or differ by an integer, at least one Frobenius solution is guaranteed; when the roots are distinct, it is usually the larger-root solution.

- 3 Otherwise x_0 is an *irregular singular point*, and the equation may have no series solution at all.

Examples: reading off P and Q at $x = 0$

Airy, $y'' - xy = 0$. Here $P(x) = 0$ and $Q(x) = -x$ are both analytic, so $x = 0$ is an *ordinary point*: two Taylor-series solutions, as we found.

$x^4 y'' + 2x^3 y' + y = 0$. Divide by x^4 : $y'' + \frac{2}{x} y' + \frac{1}{x^4} y = 0$, so $P(x) = 2/x$ is a *simple* pole (order 1) and $Q(x) = 1/x^4$ has a pole of order 4. The order-4 pole exceeds the double-pole limit, so $x = 0$ is an *irregular singular point*: no Frobenius power-series solution at the origin. Indeed the true solution is

$$y(x) = c_1 \cos\left(\frac{1}{x}\right) + c_2 \sin\left(\frac{1}{x}\right),$$

which has an essential singularity at the origin.

Examples: reading off P and Q at $x = 0$ (cont.)

$x^2 y'' - 6y = 0$. Divide by x^2 : $y'' + 0 \cdot y' - \frac{6}{x^2} y = 0$, so $x = 0$ is singular, $P(x) = 0$ is analytic, and $Q(x) = -6/x^2$ has a second-order pole. This meets Fuchs' theorem, so $x = 0$ is a *regular singular point* and we get at least one solution. Substituting $y = \sum a_k x^{k+r}$ into $x^2 y'' - 6y = 0$:

$$\sum_{k=0}^{\infty} [(k+r)(k+r-1) - 6] a_k x^{k+r} = 0 \implies [(k+r)(k+r-1) - 6] a_k = 0.$$

At $k = 0$ the indicial equation is

$$r(r-1) - 6 = 0 \implies r = 3 \text{ or } r = -2.$$

This is *not* a recurrence (no a_{k-2} couples the coefficients): from a_0 we cannot generate $a_1, a_2, \dots$. But both indicial roots give a solution, $y = x^3$ and $y = x^{-2}$, so

$$y(x) = c_1 x^3 + \frac{c_2}{x^2}.$$

Examples: reading off P and Q at $x = 0$ (cont.)

$x^3 y'' - 6y = 0$. Now $Q(x) = -6/x^3$ has a *third-order* pole, beyond Fuchs' limit. We should not expect a Frobenius power series at the origin. Substituting $y = \sum a_k x^{k+r}$ into $x^3 y'' - 6y = 0$ gives

$$\sum_{k=0}^{\infty} (k+r)(k+r-1) a_k x^{k+r+1} - 6 \sum_{k=0}^{\infty} a_k x^{k+r} = 0.$$

The lowest power x^r comes only from the second sum, so its coefficient is the indicial equation $-6a_0 = 0$. Since $a_0 \neq 0$ this is impossible: there are no Frobenius solutions of the assumed form.

Bessel, $x^2 y'' + xy' + (x^2 - m^2)y = 0$. Divide by x^2 :

$$y'' + \frac{1}{x} y' + \left(1 - \frac{m^2}{x^2}\right) y = 0.$$

So $P(x) = 1/x$ has a simple pole and Q has at most a double pole; Fuchs' theorem gives at least one Frobenius solution. For integer $m \geq 0$ this is J_m , while the second solution is not pure Frobenius; noninteger root gaps give two Frobenius solutions.

Summary: when Frobenius works

For $y'' + P(x)y' + Q(x)y = 0$ at the point x_0 :

- P, Q both analytic $\Rightarrow$ **two** analytic (series) solutions centered at x_0 .
- P at most a **simple** pole and Q at most a **double** pole (regular singular point) $\Rightarrow$ **at least one** series solution.
- Otherwise (irregular singular point) there is no general Frobenius guarantee; the method may fail.

When only one solution is guaranteed, the second must be found another way — the subject of the next section.

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- 3 Second solutions

Reduction of order: building the second solution

Frobenius gave only one solution to Bessel's equation, but a second-order equation has *two* independent solutions, just as $y'' + y = 0$ has $a_0 \cos x$ and $a_1 \sin x$. Suppose $y(x)$ and $v(x)$ both solve $y'' + Py' + Qy = 0$:

$$(1) \quad y'' + Py' + Qy = 0, \quad (2) \quad v'' + Pv' + Qv = 0.$$

Form $v \cdot (1) - y \cdot (2)$; the two Q terms give $Q(vy - yv) = 0$ and drop out, leaving

$$(vy'' - yv'') + P(vy' - yv') = 0.$$

Notice that $(vy' - yv')' = v'y' + vy'' - y'v' - yv'' = vy'' - yv''$ (the $v'y'$ and $y'v'$ cross terms cancel). Writing $W = vy' - yv'$ (the Wronskian-type combination), the first group is exactly W' :

$$W' + P W = 0.$$

Reduction of order: building the second solution (cont.)

This is first order. Multiply by the integrating factor $\mu(x) = \exp\left(\int_a^x P(t) dt\right)$, whose derivative is $\mu' = P\mu$. Then $\mu W' + P\mu W = \mu W' + \mu' W = (\mu W)'$, so $\mu(W' + PW) = 0$ collapses to a total derivative:

$$\frac{d}{dx} \left[\exp\left(\int_a^x P(t) dt\right) (vy' - yv') \right] = 0.$$

Hence the bracket is a constant B :

$$\exp\left(\int_a^x P(t) dt\right) (vy' - yv') = B.$$

On an interval where v is nonzero, solve for the combination and divide by v^2 :

$$\frac{vy' - yv'}{v^2} = \frac{B \exp\left(-\int_a^x P(t) dt\right)}{v^2}.$$

Reduction of order: building the second solution (cont.)

The left side is an exact derivative, $\frac{d}{dx} \left(\frac{y}{v} \right)$. Integrating gives

$y/v = C + B \int^x \exp(-\int_a^s P(t) dt) / v^2(s) ds$. The term $Cv(x)$ is just the solution we already know, so set $C = 0$ to isolate the *second solution* from the first one v :

$$y_2(x) = B v(x) \int^x \frac{\exp(-\int_a^s P(t) dt)}{v^2(s)} ds. \quad (4.3.1)$$

Two examples of reduction of order

Harmonic oscillator, $y'' + y = 0$. We already know $\cos x$ and $\sin x$, but let us recover $\sin x$ from $\cos x$ using (4.3.1). Work on any interval avoiding the zeros of $\cos x$, for example $|x| < \pi/2$. Here $P(x) = 0$ and $v(x) = \cos x$, so the exponential factor is 1:

$$y(x) = B \cos x \int_0^x \frac{1}{\cos^2 s} ds = B \cos x \tan x = B \sin x.$$

The reduction formula reproduces the expected partner.

Two examples of reduction of order (cont.)

Bessel's equation, $x^2 y'' + xy' + (x^2 - m^2)y = 0$. Here $P(x) = 1/x$ and the known solution is $v(x) = J_m(x)$. The exponential factor is

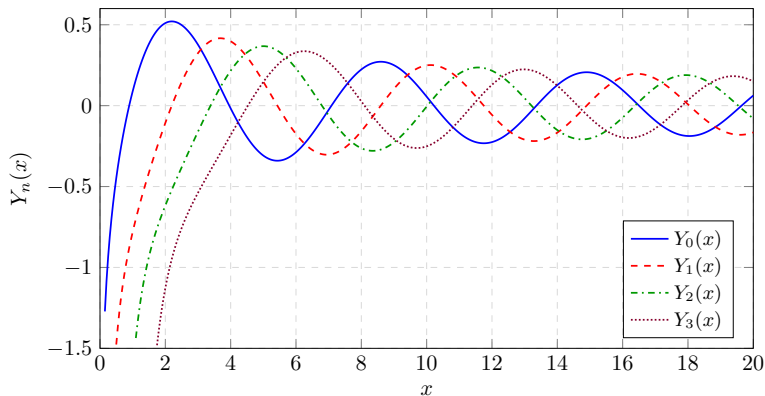
$$\exp\left(-\int_1^x t^{-1} dt\right) = \exp(-\ln x) = x^{-1}.$$

Substituting into (4.3.1),

$$y(x) = B J_m(x) \int_{x_*}^x \frac{1}{s J_m^2(s)} ds, \quad (4.3.2)$$

where $x_* > 0$ and the interval avoids zeros of J_m . This gives a second independent solution. After the conventional normalization and after adding any harmless multiple of J_m , this solution is the **Bessel function of the second kind**, denoted $Y_m(x)$. For $m > 0$, $J_m(0) = 0$ and the integrand is strongly singular; for $m = 0$, $J_0(0) = 1$ and the $1/s$ factor gives the logarithmic singularity. Thus Y_m is singular at the origin — the second, independent solution that Frobenius alone could not reach.

The Bessel functions of the second kind $Y_0, \dots, Y_3$



Unlike J_m , each Y_m plunges to $-\infty$ as $x \rightarrow 0^+$: the second Bessel solution is singular at the center, so it is excluded for a drum (finite there) but needed on annular domains.

Chapter summary

- The **method of Frobenius** solves linear second-order homogeneous ODEs by a generalized power series $y = \sum_k a_k x^{k+r}$.
- The lowest power gives the **indicial equation** for r ; matching higher powers gives a **recurrence** for the a_k .
- The harmonic oscillator yields $\cos x, \sin x$; Bessel's equation yields the **Bessel function of the first kind** $J_m(x) = \sum_k \frac{(-1)^k}{k! (m+k)!} (x/2)^{2k+m}$; Airy's equation yields multifactorial series and the integral $Ai(x), Bi(x)$.
- The other Bessel root gives $J_{-\nu}$ for noninteger order; for integer order it does not provide a new independent solution, which is why Y_m is needed.
- **Fuchs' theorem**: ordinary point $\Rightarrow$ two series solutions; regular singular point (P simple pole, Q double pole) $\Rightarrow$ at least one; irregular singular point $\Rightarrow$ possibly none.
- **Reduction of order** recovers the missing second solution:
 $y = B \int \exp(-\int P) / v^2 dx$, giving e.g. the Bessel function of the second kind Y_m .